```
C:\Windows\system32>python --version
Python 3.14.6

C:\Windows\system32>python
Python 3.14.6 (tags/v3.14.6:t63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>> print("HELLO")
HELLO
>>> exit()
```

EXPERIMENT - 4

PACKET SNIFFING USING RAW SOCKETS IN PYTHON

Aim:

To implement a packet sniffer using raw sockets in Python that captures live network traffic and parses Ethernet, IP and transport-layer (TCP/UDP/ICMP) headers.

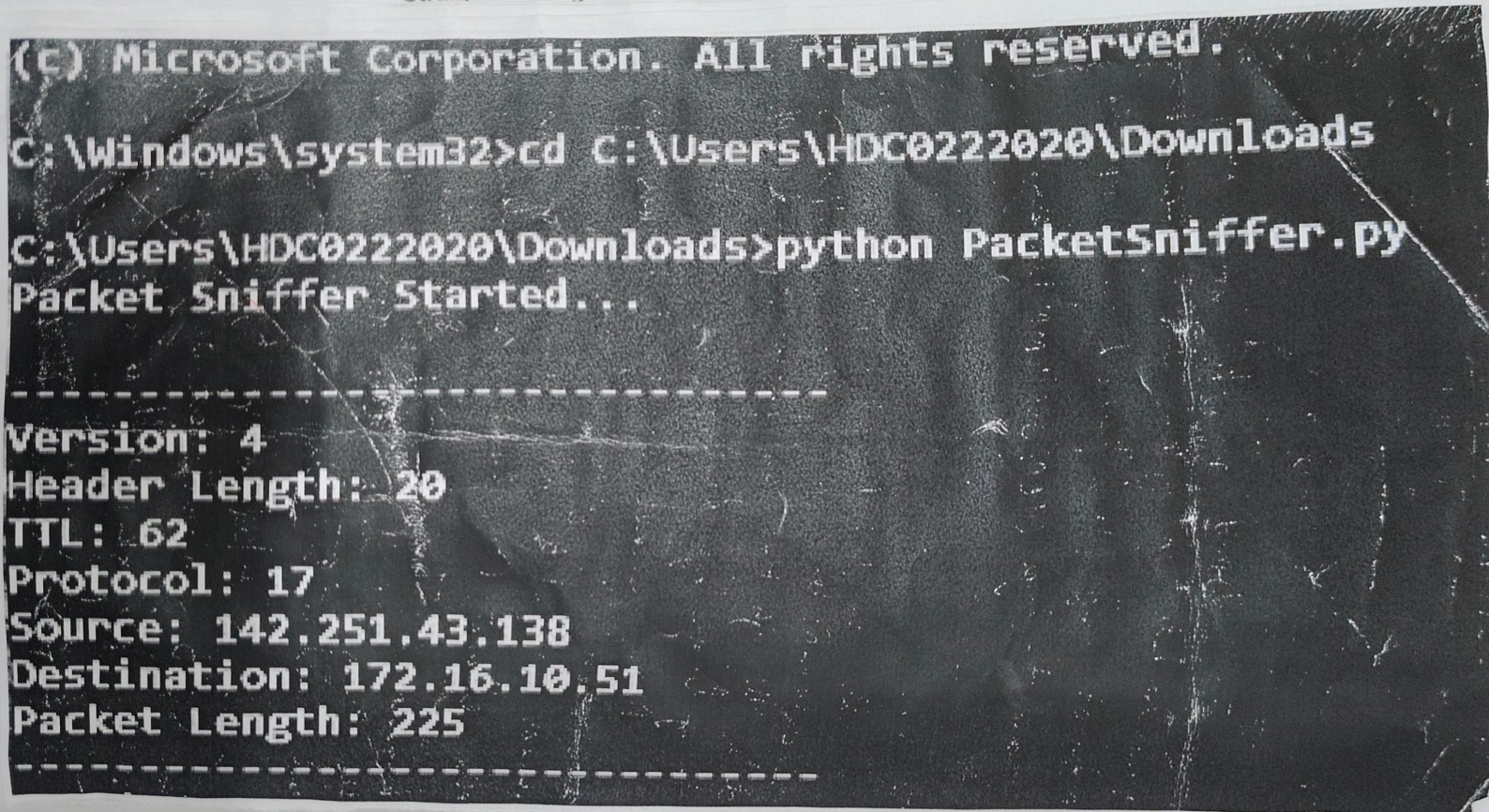
THEORY:

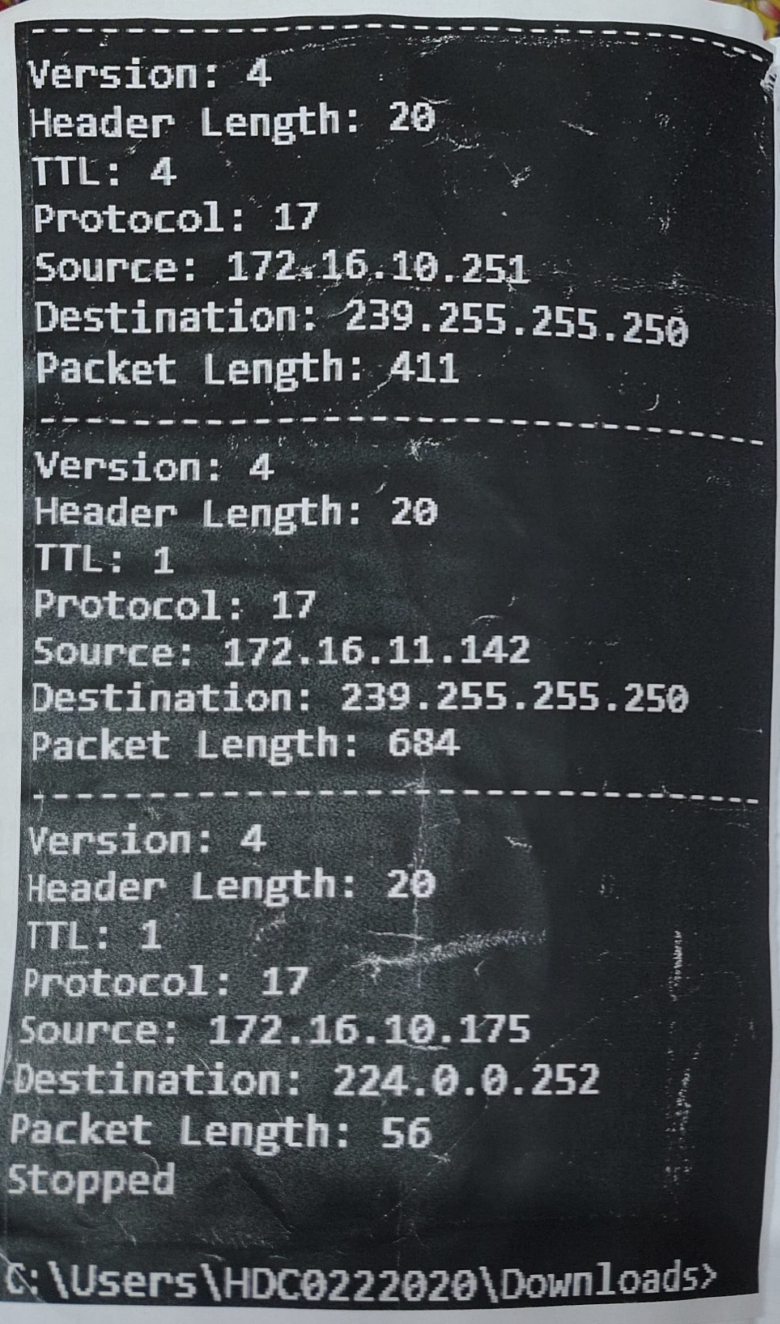
Raw Sockets: Normal sockets (SOCK_STREAM, SOCK_DGRAM) let an application send/receive data through the OS's protocol stack (TCP/UDP), hiding lower-layer headers.

A raw socket (SOCK_RAW) bypass this abstraction - it gives the program direct access to packets at the IP or even the ethernet (data link) layer, including all headers. This is why raw sockets are the basis of tools like wireshark, tcpdump and custom sniffers.

Why root/admin privileges are required:
Raw Sockets can read/inject arbitrary traffic on the network, which is a security-sensitive capability.

Straight-Through Cable (568B – 568B)





RESULT:

Packet Sniffering using raw sockets in python is completed and verified successfully.